Zeyu Zhang

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Education

University of Chinese Academy of Sciences (UCAS) Beijing, China
B.Eng. in Artificial Intelligence | GPA: 3.91/4.00 | Rank: 6/68 Sep. 2024 –Expected Jun. 2028

- **Selected Coursework:** Data Structures and Algorithms, Probability and Statistics, Discrete Mathematics, Signals and Systems, Principles of Artificial Intelligence, Pattern Recognition and Machine Learning

Research Experience

Neural Representation Learning for Embodied Control Jul. 2026 – Present
Institute of Automation, CAS | Mindtrix Technology Advisors: Bing Liu, Yinghao Li, Weihsien Lee

- Developing a neural encoder to extract behavior-relevant representations from pre-movement M1/PMd spiking activity for conditioning embodied control policies.
- Established within-session Center-Out decoding baselines and evaluated neural representations across cortical regions, temporal windows, and model architectures.
- Built an eight-direction Center-Out manipulation environment in MuJoCo and developed a fine-tuning and evaluation pipeline for the $\pi_{0.5}$ vision-language-action model.

Computer Vision Pipeline for Visual Prostheses Jul. 2025 – Aug. 2025
Institute of Automation, CAS | Mindtrix Technology Advisors: Bing Liu, Xing Li

- Reviewed depth estimation, semantic segmentation, visual representation, and neural decoding methods for visual prostheses; assisted with RealSense-based experimental data collection.
- Prototyped a vision-to-phosphene pipeline using computer vision models and dynaphos.

Publications and Manuscripts

1. Yinghao li, Weihsien lee, **Zeyu Zhang**, Canyang Zhao, Bing Liu. "NI-ACT: A Neural Intent Interface for Brain-Embodied Manipulation" *Manuscript*, 2026.

Skills

- **Technical:** Python, C, PyTorch, Git, Linux, Shell, MuJoCo, robosuite, LeRobot
- **Languages:** Chinese (native), English (TOEFL: 5.5, Duolingo: 140)

Scholarships and Awards

- First-class scholarship for undergraduate students (top 5%) 2025

Leadership & Activities

- **President, Zhiyuan Guitar Club; Co-founder and President, UCASGA** Sep. 2025–Jul. 2026
  Led the merger of two student music organizations and operations of two rehearsal spaces; organized 16 live shows, eight acoustic performances, School Live Bar, and the Summer Music Festival.
- **Venue Operations Team Lead, UCAS Grassland Music Festival** Apr. 2026–Jun. 2026
  Led venue operations, backstage logistics, and event coordination for a campus music festival.